

Spectro-Temporal Fusion of High-Gamma and Low-Frequency ECoG Signals for Intracranial Finger Movement Decoding

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Abstract—Electrocorticography (ECoG) provides a minimally invasive approach for acquiring high-fidelity intracranial brain signals. The capability to decode continuous finger movements from ECoG holds substantial promise for developing dexterous neuroprostheses, which can restore activities of daily living for individuals with motor disabilities. Despite this potential, the inherent complexity and distributed nature of finger somatotopy present a significant algorithmic challenge. To this end, we propose a novel decoding framework, driven by the specifically engineered HiLoFuseNet architecture, to effectively fuse the spectro-temporal dynamics of high-gamma and low-frequency ECoG signals for continuous finger movement trajectory prediction. HiLoFuseNet significantly outperforms twelve state-of-the-art models in our open-source benchmark, attaining the highest decoding performance on the BCI Competition IV and Stanford datasets with standard ECoG grids. Beyond superior performance, the model successfully captures the distinct neurophysiological behavior of high-gamma and low-frequency ECoG activities during finger movements, all while maintaining a concise and efficient profile in both feature extraction and network architecture. Collectively, this study provides a performant and neurophysiologically sound framework for continuous finger movement decoding from ECoG signals, establishing a solid foundation for the development of mo-

tor brain-computer interfaces.

Index Terms—Brain-computer interface, ECoG, decoding, finger movement, neural network.

Code and source data:  [HiLoFuseNet](#)

I. INTRODUCTION

FINGER movements are central to activities of daily living (ADL). However, individuals with motor cortex stroke, spinal cord injury, or amyotrophic lateral sclerosis often experience a loss of hand mobility, which severely diminishes their independence in ADL. Neurotechnologies hold the potential to substitute or restore damaged hand motor or sensory function [1]. Particularly, Brain-Computer Interface (BCI)-based neuroprostheses provide a direct communication pathway between the brain and external devices [2], bypassing defective neural pathways to enable hand control and functional restoration [3]–[5]. Motor decoding from brain signals constitutes a critical module of such BCI systems.

The performance of hand motor decoding, specifically in terms of achievable degrees of freedom (DoFs), depends on the employed neural recording technique [1]. Non-invasive electroencephalography (EEG) studies have demonstrated the feasibility of decoding discrete hand grasping [7], [8]. However, decoding individual finger movements remains challenging due to the limited spatial resolution and low signal-to-noise ratio (SNR) of EEG [9]. Conversely, intracortical techniques using microelectrode arrays (MEA) have demonstrated high-performance continuous finger movement decoding in both non-human primates [4], [10] and paralyzed human participants [11]. Electrocorticography (ECoG) stands as an intermediate choice between EEG and MEA recordings. By recording brain activity via epidural or subdural cortical electrodes, ECoG provides a trade-off between SNR and spatial resolution [12]. Leveraging these advantages, ECoG signals enable the reliable decoding of a range of discrete hand postures [13]–[15] and individual finger movements [16]. Recent studies are actively pushing the limits of ECoG applications towards continuous finger movement (kinematics) decoding [17]–[24].

Decoding finger movements from ECoG remains challenging because the representation of individual fingers in the primary motor cortex is not fully segregated, but instead exhibits functional predominance within a highly distributed

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TABLE I
SUMMARY OF RECENT ECoG STUDIES FOCUSED ON HAND AND FINGER MOVEMENT DECODING (2016–PRESENT)

Study	Task	Feature	Model
Hotson et al., 2016 [16]	Finger Cls.	Power of gamma-band (72–110 Hz).	LDA
Saa et al., 2016 [25]	Finger Cls.	Analytic amplitude of high gamma (70–170 Hz).	Conditional Random Field
Eliseyev et al., 2016 [26]	Hand Reg.	Morlet wavelet amplitudes (15 frequencies, 10–150 Hz, linear scale).	Penalized NPLS
Schaeffer and Aksenova, 2016 [27]	Hand Reg.	Morlet wavelet amplitudes (38 frequencies, 1–250 Hz, logarithmic scale).	Markov Switching Linear Model
Flint et al., 2016 [28]	Hand/finger Reg.	Multi-band power (5 bands, including 0–4 Hz, 7–20 Hz, 70–115 Hz, 130–200 Hz, 200–300 Hz) and LMPs.	Wiener cascade decoder
Branco et al., 2017 [15]	Gesture Cls.	Power of gamma-band (70–125 Hz).	Discriminate Spatial Patterns
Xie et al., 2018 [19]	Finger Reg.	Preprocessed ECoG signals.	CNN-LSTM
Petrosyan et al., 2021 [29]	Finger Reg.	Preprocessed ECoG signals.	CNN
Yao et al., 2022 [20]	Finger Cls./Reg.	Multi-band power (5 bands, including alpha (8–13 Hz), beta (13–30 Hz), low-gamma (30–60 Hz), gamma (60–100 Hz), and high-gamma (100–200 Hz)), LMPs, Hjorth Parameters, and Riemannian features.	LightGBM
Sliwowski et al., 2022 [21]	Hand Reg.	Morlet wavelet amplitudes (15 frequencies, 10–150 Hz, linear scale).	CNN-LSTM
Faes et al., 2022 [22]	Finger Reg.	Spectral amplitudes in delta (1.5–5 Hz), theta (5–8 Hz), alpha (8–12 Hz), beta1 (12–24 Hz), beta2 (24–34 Hz), gamma1 (34–60 Hz), gamma2 (60–100 Hz), and gamma3 (100–130 Hz).	Block-Term Tensor Regression (BTTR)
Faes and Van Hulle, 2022 [23]	Finger Reg.	Spectral amplitudes in delta (1.5–5 Hz), theta (5–8 Hz), alpha (8–12 Hz), beta1 (12–24 Hz), beta2 (24–34 Hz), gamma1 (34–60 Hz), gamma2 (60–100 Hz), and gamma3 (100–130 Hz).	Extended BTTR
Merino et al., 2023 [30]	Finger Reg.	LMPs and high gamma band power (100–200 Hz).	Ordinary Least Squares
Faes et al., 2025 [24]	Finger Reg.	LMP power (0.1–1.5 Hz) and gamma amplitudes (15 overlapping 10-Hz bands from 60–130 Hz).	Graph-optimized BTTR
Ferdinand et al., 2025 [31]	Hand Reg.	Morlet wavelet amplitudes (15 frequencies, 10–150 Hz, linear scale).	Transformer
Lin et al., 2025 [32]	Finger Reg.	Morlet wavelet power (20 frequencies, 5–195 Hz, logarithmic scale).	CNN-BiLSTM, RF
Tao et al., 2025 [33]	Finger Reg.	Morlet wavelet power (40 frequencies, 40–300 Hz, logarithmic scale).	DeepFingerNet

1) Abbreviations: Cls., classification; Reg., regression; LDA, Linear Discriminant Analysis; LMPs, local motor potentials; NPLS, N-way partial least squares; GBM, gradient boosting machine; CNN, convolutional neural network; (Bi)LSTM, (bidirectional) long short-term memory, RF: random forest.

2) The table focuses on ECoG decoding studies of hand and finger movements. Studies focusing primarily on proximal upper-limb or arm-reaching movements are not included.

and overlapping neural network [34], [35]. To address this physiological complexity, recent studies have focused on better feature engineering [20], [30] and advanced machine learning (ML) [20], [22], [26] and deep learning models [19], [21], [31], [32]. Table I summarizes representative ECoG studies reported over the past decade for hand and finger movement decoding.

Despite this progress, two fundamental challenges remain unresolved. First, feature extraction for ECoG decoding is still highly empirical and computationally demanding. Existing studies employ diverse signal representations, ranging from local motor potentials and multi-band spectral power to dense wavelet decompositions across broad frequency ranges, making it difficult to establish a unified and computationally efficient feature extraction strategy. Although high-gamma activity is widely recognized as one of the most informative neural correlates of movement, its frequency boundaries, extraction protocols, and integration with complementary low-frequency information vary substantially across studies. Second, the practical advantage of deep neural networks (DNNs) for continuous finger movement decoding remains unclear. While DNNs offer strong representation learning capabilities for modeling nonlinear spatiotemporal dynamics, recent evidence suggests that they do not consistently outperform carefully designed conventional ML approaches in ECoG decoding benchmarks [20], [22], [23].

To address these limitations, this study develops a performant and neurophysiologically grounded framework for continuous finger movement decoding from ECoG. Our approach includes two complementary strategies. First, we propose a computationally concise signal processing pipeline that minimizes feature engineering complexity while preserving the

most informative neural components. Second, we design an efficient hybrid DNN architecture that effectively exploits the complementary spatiotemporal characteristics of ECoG signals and achieves state-of-the-art decoding performance while maintaining physiological interpretability. The specific contributions of this study are summarized as follows:

- 1) We propose a streamlined ECoG feature extraction pipeline for continuous finger movement decoding that efficiently captures complementary high-gamma activity (HGA) and low-frequency signals (LFS), substantially reducing the complexity of conventional multi-band feature engineering.
- 2) We introduce **High-gamma and Low-frequency ECoG signal Fusion Network (HiLoFuseNet)**, a novel hybrid DNN architecture that effectively integrates HGA and LFS through a depthwise separable convolution and LSTM backbone, delivering superior decoding performance across two public finger movement datasets.
- 3) We establish a standardized benchmarking framework comprising twelve representative baselines, under which HiLoFuseNet achieves the best reproducible decoding performance and supports the potential of DNN-based approaches for continuous finger movement decoding.
- 4) We demonstrate that HiLoFuseNet learns neurophysiologically meaningful representations, attending to focal sensorimotor regions for HGA while exploiting more distributed cortical dynamics for LFS, and we release the complete benchmark code to facilitate reproducible research.

II. RELATED WORKS

Continuous decoding of finger movements from human ECoG signals was first demonstrated by Kubanek et al. [17],

who extracted spectral features across multiple frequency bands together with local motor potentials (LMPs), and employed a linear multivariate decoder. Their study established high-gamma activity and LMP as the most informative neural signatures for finger kinematics decoding. Building upon this foundation, subsequent studies have mainly followed two methodological directions: tensor regression models based on Partial Least-Squares (PLS) and deep neural networks (DNNs) (cf. Table I).

The first line of work focuses on exploiting the multi-dimensional structure of ECoG data through PLS-based regression. Shimoda *et al.* [36] applied conventional PLS with wavelet features to decode continuous 3D hand trajectories. To preserve the inherent tensor structure of neural data, higher-order extensions were subsequently introduced. Zhao *et al.* [37] proposed Higher-Order PLS (HOPLS), which demonstrated superior performance over both standard PLS and N-way PLS (NPLS) for trajectory decoding. Later, Camarrone *et al.* [38] and Faes *et al.* [22], [23] developed Block-Term Tensor Regression (BTTR) and its extended variant (eBTTR), further improving flexibility in modeling multi-DoF finger movements through block-wise tensor decomposition.

In parallel, DNN-based approaches have emerged as a powerful alternative by jointly learning spatial, spectral, and temporal representations directly from neural data. Xie *et al.* [19] first introduced an end-to-end CNN-LSTM architecture for continuous finger decoding, achieving better performance than the BCI Competition IV winning linear decoder [18]. Subsequent studies explored more compact CNN models [29] and hybrid CNN-BiLSTM architectures [32], demonstrating strong decoding capability. More recently, Tao *et al.* [33] reported state-of-the-art performance using a CNN-based U-Net architecture on wide-band wavelet representations. In parallel, Ferdinand *et al.* [31] introduced Transformer-based models for hand position decoding, demonstrating promising performance in continuous movement regression tasks.

III. HILOFUSENET DECODING FRAMEWORK

We propose HiLoFuseNet, a compact decoding framework that jointly exploits complementary spectro-temporal information from high-gamma and low-frequency ECoG signals for continuous finger movement decoding. As illustrated in Fig. 1, the framework consists of two key components: (a) parallel feature extraction and (b) a lightweight neural decoder.

A. Parallel Feature Extraction

To capture complementary neural dynamics across frequency bands, each preprocessed ECoG segment is transformed into two parallel feature streams: High-Gamma Activity (HGA) and Low-Frequency Signals (LFS). Let $\mathbf{X} \in \mathbb{R}^{C \times T_0}$ denote an ECoG segment from the continuous signal, where C is the number of channels and T_0 is the number of temporal samples. As shown in Fig. 1(a), $\mathbf{X}$ is processed by two parallel streams to construct a unified spectro-temporal representation.

The upper stream extracts HGA features. First, the signals are bandpass filtered within 70–200 Hz using a causal 4th-order Butterworth IIR filter. The analytic amplitude is then

obtained via the Hilbert transform, followed by temporal decimation through average pooling over every five consecutive samples. This produces an HGA feature sequence sampled at 200 Hz¹:

$$\mathbf{X}^{\text{HGA}} = \text{AvgPool}(|\mathcal{H}(\mathbf{X}^{70-200 \text{ Hz}})|) \in \mathbb{R}^{C \times T}, \quad (1)$$

where:

- $\mathbf{X}^{70-200 \text{ Hz}}$ denotes the bandpass-filtered ECoG signal;
- $|\mathcal{H}(\cdot)|$ denotes the amplitude envelope of the analytic signal, obtained by applying the Hilbert transform followed by the modulus operation.
- $\text{AvgPool}(\cdot)$ denotes average pooling for temporal decimation;
- T is the temporal dimension after downsampling.

In parallel, the lower stream extracts LFS features by first applying an anti-aliasing low-pass filter (a causal 4th-order Butterworth IIR filter) with a cutoff frequency of 100 Hz, followed by downsampling to the same 200 Hz sampling rate:

$$\mathbf{X}^{\text{LFS}} = \text{Downsample}(\mathbf{X}^{\text{FilterLP}}) \in \mathbb{R}^{C \times T}, \quad (2)$$

where $\mathbf{X}^{\text{FilterLP}}$ denotes the low-pass filtered signal.

Finally, the extracted HGA and LFS features are stacked along the frequency dimension to form the spectro-temporal representation:

$$\mathbf{X}^{\text{spec}} = \text{Stack}(\mathbf{X}^{\text{HGA}}, \mathbf{X}^{\text{LFS}}) \in \mathbb{R}^{F \times C \times T}, \quad (3)$$

where $F = 2$ corresponds to the HGA and LFS feature streams.

This two-stream feature extraction strategy is inspired by the parallel spectro-temporal representation proposed by Metzger *et al.* [39]. In contrast to the latter, our design simplifies HGA extraction and reformulates the feature representation for efficient continuous finger trajectory decoding.

B. Network Architecture

As illustrated in Fig. 1(b), the proposed neural decoder consists of three modules: (1) depthwise separable convolution (DS-Conv), (2) LSTM-based temporal modeling, and (3) a multilayer perceptron (MLP) regression head. Detailed layer configurations are summarized in Supplementary Table S1. For clarity, the batch dimension is omitted throughout this section.

1) DS-Conv Module: The DS-Conv module extracts compact spectro-temporal features from the input representation $\mathbf{X}^{\text{spec}}$. Following the design of EEGNet [40], a depthwise spatial convolution with kernel size $(C, 1)$ is first applied independently to each spectral component, followed by Batch Normalization (BN) and Exponential Linear Unit (ELU) activation:

$$\mathbf{Z}_1 = \text{ELU}(\text{BN}(\text{Conv}_{\text{DW}, \mathbf{W}_s}(\mathbf{X}^{\text{spec}}))) \in \mathbb{R}^{(F \cdot D) \times 1 \times T}, \quad (4)$$

where $\mathbf{W}_s \in \mathbb{R}^{D \times C}$ denotes the D number of spatial convolution kernels for each stream F .

¹This implementation assumes that the original ECoG sampling rate exceeds 200 Hz. Otherwise, both the high-gamma cutoff frequency and target sampling rate should be adjusted accordingly. In this study, all datasets were sampled at 1000 Hz.

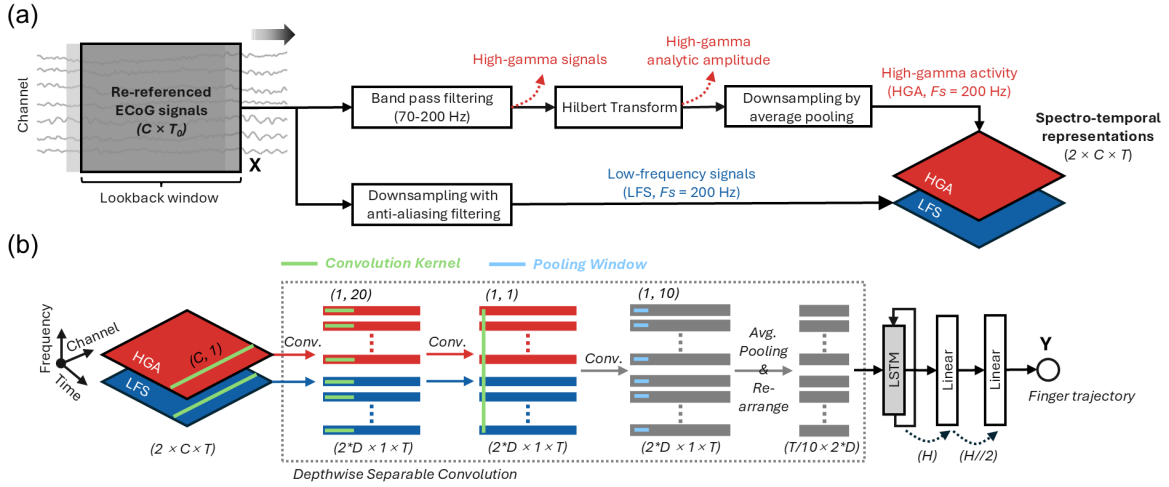


Fig. 1. Illustration of the **High-gamma and Low-frequency ECoG signal Fusion Network (HiLoFuseNet)**. (a) Pipeline to extract high-gamma activity (HGA) and low-frequency signals (LFS) from a segment of ECoG signals. (b) The network architecture combining depthwise separable convolution and LSTM for learning to extract spectro-temporal information. C denotes the number of ECoG channels, T_0 and T the number of temporal samples. D and H are model hyper-parameters that control the number of CNN spatial filters and the hidden states of LSTM.

Next, depthwise temporal convolution with kernel size (1, 20), corresponding to 100 ms window, is applied to capture local temporal patterns, followed by a pointwise convolution for cross-feature integration:

$$\mathbf{Z}_2 = \text{ELU}(\text{BN}(\text{Conv}_{\text{PW}, \mathbf{W}_p} \text{Conv}_{\text{DW}, \mathbf{W}_t}(\mathbf{Z}_1))) \in \mathbb{R}^{(F \cdot D) \times 1 \times T}, \quad (5)$$

where $\mathbf{W}_t$ and $\mathbf{W}_p$ denote the temporal and pointwise convolution kernels, respectively.

Finally, average pooling (kernel size of (1, 10), corresponding to a 50-millisecond temporal window) and dropout are applied for temporal compression and regularization:

$$\mathbf{Z}_{\text{CNN}} = \text{Dropout}(\text{AvgPool}(\mathbf{Z}_2)) \in \mathbb{R}^{(F \cdot D) \times 1 \times T/10}. \quad (6)$$

2) LSTM Module: The convolutional features are reshaped into a temporal sequence $\mathbf{Z}'_{\text{CNN}} \in \mathbb{R}^{T/10 \times (F \cdot D)}$ and fed into a single-layer LSTM to model long-range temporal dependencies:

$$(\mathbf{h}_t, \mathbf{c}_t) = \text{LSTM}(\mathbf{z}'_{\text{CNN}, t}, \mathbf{h}_{t-1}, \mathbf{c}_{t-1}), t = 1, 2, \dots, T/10, \quad (7)$$

where $\mathbf{h}_t$ and $\mathbf{c}_t$ denote the hidden state and cell state at sequence step t , respectively. The final hidden state $\mathbf{h}_{\text{final}} \in \mathbb{R}^H$ is used as the sequence representation.

3) Output Module: The final hidden state is passed through a two-layer MLP to generate finger trajectory predictions:

$$\mathbf{Y} = (\text{Dropout}(\text{ReLU}(\mathbf{h}_{\text{final}} \mathbf{W}_1 + \mathbf{b}_1))) \mathbf{W}_2 + \mathbf{b}_2 \in \mathbb{R}^O, \quad (8)$$

where $\mathbf{W}_1$, $\mathbf{W}_2$, $\mathbf{b}_1$, and $\mathbf{b}_2$ denote the learnable parameters of the MLP, and O is the output dimension. $O = 5$ in this study for simultaneous five-finger trajectory decoding.

IV. MATERIALS AND METHODS

A. ECoG Datasets

The proposed model was evaluated on two publicly available ECoG datasets for finger movement decoding. The first

is the BCI Competition IV dataset (denoted as *BCIIV*), which includes three subjects [41]. The second is the *fingerflex* dataset from the Stanford Human ECoG Library (denoted as *Stanford*), which includes nine subjects [42]². All participants were epilepsy patients implanted with subdural ECoG electrodes for clinical monitoring. Both datasets employed an identical finger movement paradigm. In each trial, a visual cue instructed one of five fingers, followed by a 2-s self-paced movement period and a 2-s rest period. Each finger was repeated for 30 trials in randomized order. During the experiment, ECoG signals and five finger trajectories, recorded using a sensorized data glove, were simultaneously acquired. For both datasets, ECoG signals were sampled at 1000 Hz, while finger trajectories were sampled at 25 Hz. Instrumental band-pass filtering was set to 0.15-200 Hz for both datasets. Detailed dataset information is provided in Supplementary Table S2.

B. Data Preprocessing and Preparation

Raw continuous ECoG signals were first notch-filtered to suppress power-line interference (60 Hz) and its second and third harmonics using a one-pass 3rd-order Butterworth IIR filter. Channels exhibiting abnormal activity were subsequently identified through visual inspection and excluded; details of the rejected channels are provided in Supplementary Table S2. The remaining channels were then re-referenced using common average referencing (CAR). All preprocessing steps were implemented causally to ensure compatibility with real-time decoding.

The BCIIV dataset provides a predefined split of 400 s for training and 200 s for testing, which was retained for

²Ethics statement: All patients participated in a purely voluntary manner, after providing informed written consent, under experimental protocols approved by the Institutional Review Board of the University of Washington (#12193). All patient data was anonymized according to IRB protocol, in accordance with HIPAA mandate. These data originally appeared in the manuscript "Human Motor Cortical Activity Is Selectively Phase-Entrained on Underlying Rhythms" published in PLoS Computational Biology in 2012 [43].

fair comparison with previous studies. The Stanford dataset consists of continuous recordings and was partitioned following [32]: recordings longer than 600 s were split into the first 400 s for training and the remaining data for testing; otherwise, recordings were divided into training and testing sets with a 2:1 ratio. To construct training and testing samples, a sliding window was applied to the continuous signals, where each input segment preceded the target movement to preserve causality. The window stride was set to 40 ms, matching the 25 Hz sampling interval of the data glove. The last 10% of the training samples was held out as the validation set. Z-score normalization was applied to all datasets using statistics computed exclusively from the remaining training samples to prevent data leakage.

C. Compared Methods

We compare HiLoFuseNet with representative state-of-the-art decoding methods. As previous studies employed different preprocessing and train-test splitting protocols, all baseline methods were re-implemented as closely as possible to their original descriptions under the proposed benchmarking framework for fair comparison. The source code is publicly released with this work. Please refer to  HiLoFuseNet.

1) **PLS**: PLS regression is applied to vectorized ($channel \cdot time \cdot frequency$) Morlet wavelet amplitudes (10–150 Hz, 15 linearly spaced frequencies) following [26]. The temporal dimension is downsampled to 10 Hz. The number of latent components K is searched in [1, 50].

2) **NPLS**: NPLS regression is applied directly to tensorized ($channel \times time \times frequency$) wavelet features extracted identically to PLS [26]. The number of latent components K is searched in [1, 50].

3) **HOPLS**: HOPLS [37] is applied to the same wavelet features, reshaped as $channel \times (time \cdot frequency)$. The block number K and uniform rank λ are searched in [1, 50] and [1, 20], respectively.

4) **eBTTR**: eBTTR [23] is evaluated using spectral amplitudes from eight physiological frequency bands ($channel \times time \times frequency$). Hyperparameters follow the original work, with block number $K \in [1, 50]$, SNR $\in [1, 50]$, and $\tau \in [95, 99]$.

5) **RF**: RF regression is applied to vectorized Morlet wavelet power features (5–195 Hz, 20 logarithmically spaced frequencies) following [32]. The model consists of 200 decision trees with $\sqrt{d}$ feature sampling at each split. The grid search for the random forest hyperparameters includes maximum tree depths of 10, 20, and unlimited, along with minimum samples per leaf of 2, 5, and 10.

6) **MLP_{HGA}**: A standard three-layer MLP with ReLU activations is implemented using vectorized HGA features as input. HGA is extracted as described in Section III-A and temporally downsampled with a window size of 10. The hidden dimension is set to 256.

7) **LSTM_{HGA}**: We re-implemented a LSTM-based decoder inspired by Costello *et al.* [10], which originally used spiking-band power (SBP) features for MEA recordings. In our adaptation, SBP is replaced by HGA. HGA is temporally

downsampled with a window size of 10, and the hidden dimension is set to 256. The regression head is identical to that of HiLoFuseNet.

8) **EEGNet**: EEGNet-8,2 [40] is adapted for continuous regression to evaluate the effectiveness of the proposed DS-Conv design. The model uses preprocessed ECoG signals ($channel \times time$) as input.

9) **DeepFingerNet**: DeepFingerNet [33] is re-implemented using Morlet wavelet power features (40–200 Hz, 32 logarithmically spaced frequencies). The model adopts a U-Net architecture and takes flattened features ($(channel \cdot frequency) \times time$) as input.

10) **CNN-LSTM**: The CNN-LSTM model proposed in [32] is re-implemented using Morlet wavelet power features (5–195 Hz, 20 logarithmically spaced frequencies), temporally downsampled to 100 Hz. The resulting tensor ($channel \times time \times frequency$) is fed into the original hybrid CNN and bidirectional LSTM architecture.

11) **WaT and WaTEi**: WaT and WaTEi [31] are re-implemented using Morlet wavelet amplitudes identical to PLS. Both models employ two Transformer layers with four attention heads. The internal feature dimension is set to 256 for WaT and 128 for WaTEi.

D. Benchmarking Framework

To ensure a fair comparison across models, we construct a unified benchmarking framework, as performance in movement decoding is highly sensitive to experimental settings such as lookback window length, prediction frequency, and feature extraction causality. In addition, stochastic factors in DNNs (e.g., random initialization) and differences in optimization objectives can further confound reported performance across studies. Within this framework, all models are evaluated under identical experimental conditions. The lookback window is varied from 0.4 s to 2.8 s with a step size of 0.2 s, while the prediction frequency is fixed at 25 Hz to match the sampling rate of the finger trajectory signals. To emulate a real-time decoding scenario, a causal sliding window is applied to the continuous ECoG signals prior to feature extraction, ensuring that only past information is used at each prediction step. For DNN models, stochastic variability is further controlled by fixing the random seed to 42, a conventional arbitrary choice used to ensure reproducibility. The training objective for all DNNs is standardized to mean squared error (MSE) loss.

E. Model Training and Evaluation

All DNNs were trained using the Adam optimizer with an initial learning rate of 0.001 and MSE loss function. A *ReduceLROnPlateau* scheduler was applied to the validation Pearson correlation, reducing the learning rate by a factor of 0.5 if no improvement was observed for 5 epochs. Early stopping was also based on validation correlation with a patience of 15 epochs. Training was performed with a batch size of 128 for a maximum of 200 epochs. The model parameters corresponding to the best validation performance were restored before testing. All deep learning models were implemented in PyTorch and trained on an NVIDIA A100

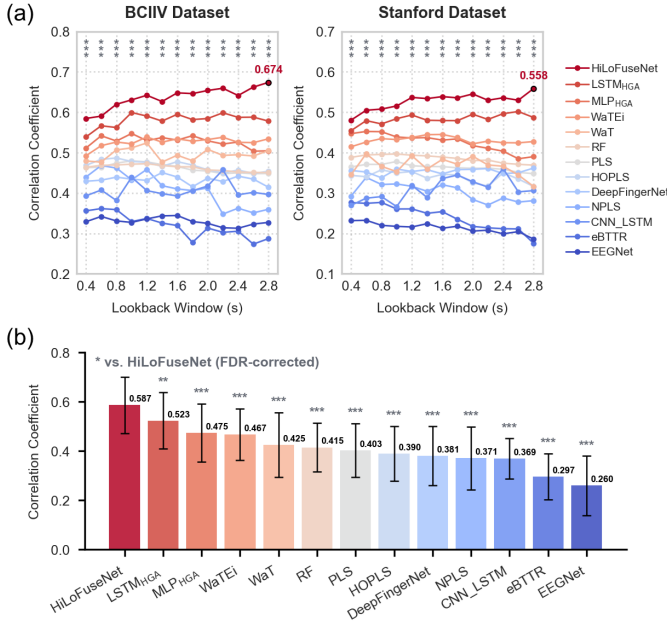


Fig. 2. Decoding performance comparison across all models. (a) Mean Pearson correlation coefficient, averaged across five fingers and all subjects, as a function of lookback window on the BCIIV and Stanford datasets. Each curve represents one decoding model, with highlighted values indicating the best overall performance. Asterisks indicate significant differences across models after Friedman test and FDR correction across lookback windows. (b) Statistical comparison of the mean decoding performance using each model's optimal lookback window. Error bars indicate standard deviation across subjects. Asterisks denote statistically significant differences compared with HiLoFuseNet after FDR correction. *** $p < 0.001$, ** $p < 0.01$.

GPU. For machine learning baselines, hyperparameters were selected using the validation set. After tuning, each model was retrained on the full training set with the optimal configuration.

Model performance was evaluated using the Pearson correlation coefficient (R) between predicted and ground-truth trajectories, consistent with prior studies [19], [20], [23], [32].

V. RESULTS AND ANALYSES

A. Decoding Performance

Fig. 2 compares the decoding performance of HiLoFuseNet against representative baseline methods on the BCIIV and Stanford datasets. For each dataset, the Pearson correlation coefficient was averaged across all five fingers and all subjects. As shown in Fig. 2(a), HiLoFuseNet consistently achieved the highest decoding accuracy across all tested lookback windows on both datasets. The Friedman test further confirmed significant model-dependent differences at each lookback window after FDR correction across windows. Performance generally improved with increasing temporal context and gradually saturated beyond approximately 2 s. The best performance of HiLoFuseNet was achieved at a lookback window of 2.8 s, reaching $R = 0.674$ on the BCIIV dataset and $R = 0.558$ on the Stanford dataset, substantially outperforming all competing methods. To provide a fair model-wise comparison, each method was evaluated using its optimal lookback window length, selected by maximizing the average decoding performance across all subjects from both datasets combined. As shown in Fig. 2(b), HiLoFuseNet achieved the highest overall

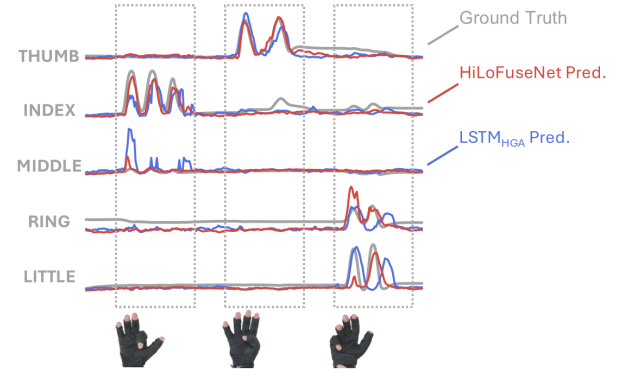


Fig. 3. Representative finger trajectory prediction on Subject 2 from the BCIIV dataset over a 12-s segment sampled at 25 Hz. Gray curves denote the ground truth, red curves denote HiLoFuseNet predictions, and blue curves denote predictions from the second-best baseline, LSTM_{HGA}. Dotted boxes highlight representative index, thumb, and little finger movement epochs, respectively.

decoding performance ($R = 0.587$), followed by LSTM_{HGA} ($R = 0.523$) and MLP_{HGA} ($R = 0.475$). Compared with the strongest baseline, HiLoFuseNet improved the decoding correlation by 10.9%. Statistical analysis further confirmed that HiLoFuseNet significantly outperformed all baseline methods after FDR correction ($p < 0.01$). The detailed per-finger and per-subject results are reported in Supplementary Tables S3 and S4.

Fig. 3 presents a representative comparison between HiLoFuseNet and the second-best baseline, LSTM_{HGA}, using a 12-s prediction segment from Subject 2 of the BCIIV dataset. Additional trajectory visualizations across subjects are provided in Supplementary Fig. S1. Overall, HiLoFuseNet more accurately reproduced the temporal profile and amplitude of the ground-truth finger trajectories across the five fingers. In the highlighted movement epochs, HiLoFuseNet tracked repeated flexion events with higher fidelity, producing peak timings and amplitudes that more closely matched the ground truth, whereas LSTM_{HGA} showed larger deviations, including missed or distorted peaks and spurious transient activations.

B. Analysis of Learned Representations

We analyzed the spatial representations learned by the DS-Conv module. This analysis was conducted on the Stanford dataset, which provides subject-specific cortical reconstructions and electrode localization required for anatomical mapping. The learned spatial kernel weights ($\mathbf{W}_s$) were normalized and projected onto the cortical surface and separately analyzed for the HGA and LFS streams.

As illustrated in Fig. 4(a), both feature streams consistently assigned high attention to sensorimotor regions associated with hand movement, particularly the dorsal M1, dorsal S1, and ventral sensorimotor cortex surrounding the anatomical hand knob. However, distinct spatial characteristics emerged between the two streams. The HGA stream exhibited a highly localized attention pattern, with dominant activation concentrated around the dorsal sensorimotor cortex. In contrast, the LFS stream showed a more spatially distributed attention pattern, extending beyond the sensorimotor cortex into broader

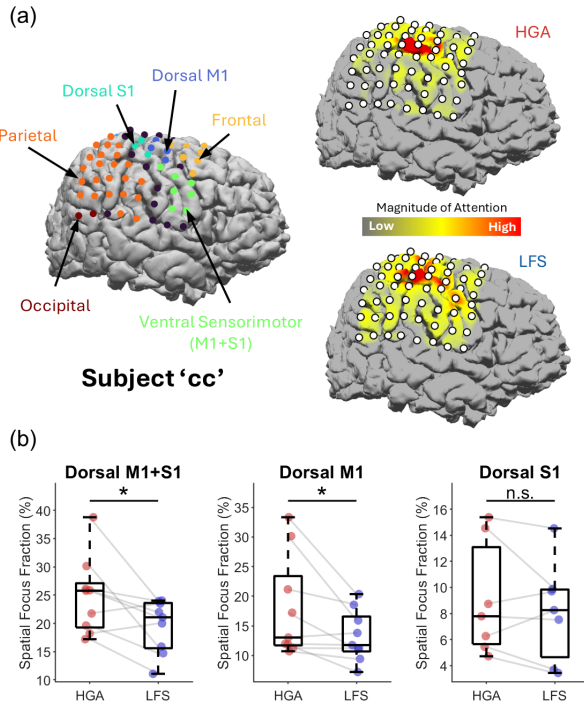


Fig. 4. Analysis of the spatial representations learned by HiLoFuseNet on the Stanford dataset. (a) Representative cortical attention maps of Subject ‘cc’ for the HGA and LFS streams, obtained by projecting the learned spatial kernel weights onto the individual cortical surface. Warmer colors indicate higher model attention. Anatomical regions are annotated. (b) Quantitative comparison of the spatial focus fraction within sensorimotor subregions across subjects. Each point represents one participant, and gray lines connect paired HGA and LFS measurements. Asterisks denote statistically significant differences ($p < 0.05$, Wilcoxon signed-rank test), while n.s. indicates no significant difference.

frontal and parietal regions. To quantitatively characterize this difference, we computed the spatial focus fraction, defined as the proportion of model attention allocated within predefined sensorimotor regions of interest. As shown in Fig. 4(b), the HGA stream allocated significantly greater attention to the dorsal sensorimotor cortex (dorsal M1+S1, $p < 0.05$) and specifically to dorsal M1 ($p < 0.05$) compared with the LFS stream, whereas no significant difference was observed in dorsal S1 alone ($p > 0.05$). These findings suggest that HGA features primarily capture focal motor-related cortical activity, while LFS features encode more distributed network-level dynamics.

This complementary spatial organization is consistent with established neurophysiology. During voluntary movement, low-frequency rhythms, particularly μ (8–12 Hz) and β (18–25 Hz), exhibit widespread event-related desynchronization across the sensorimotor cortex, whereas high-gamma activity (> 75 Hz) is known to be spatially focal and closely linked to local neuronal firing [44], [45]. Therefore, the distinct attention patterns learned by HiLoFuseNet provide physiologically interpretable evidence that the proposed dual-stream architecture successfully captures complementary neural dynamics across frequency bands.

C. Ablation Study and Hyperparameter Analysis

To quantify the contribution of each architectural component, we constructed four HiLoFuseNet variants by selec-

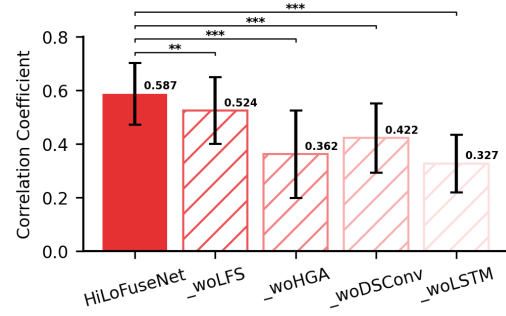


Fig. 5. Ablation analysis of HiLoFuseNet. Performance is reported as mean correlation coefficient across fingers, subjects, and datasets for the full model and four ablated variants. Error bars denote standard deviation. Asterisks indicate statistically significant performance differences between HiLoFuseNet and each ablated variant (** $p < 0.01$, *** $p < 0.001$, Wilcoxon signed-rank test with FDR correction).

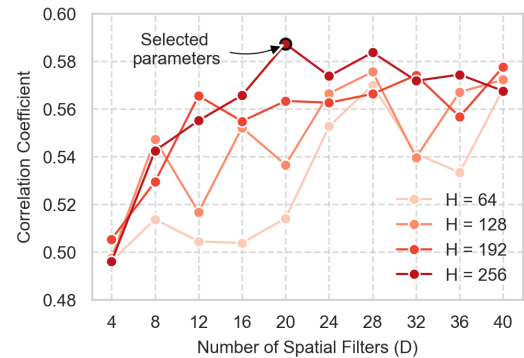


Fig. 6. Sensitivity analysis of model hyperparameters. Performance is reported as mean correlation coefficient across fingers, subjects, and datasets.

tively removing key modules: (1) _woLFS: the LFS feature stream was removed; (2) _woHGA: the HGA feature stream was removed; (3) _woDSConv: the DS-Conv module was omitted, and the spectro-temporal representations ($\mathbf{X}^{\text{spec}}$) were reshaped into sequential inputs ($\mathbb{R}^{T \times (F \cdot C)}$) for direct LSTM processing; (4) _woLSTM: the temporal modeling module was removed, and the flattened DS-Conv outputs were directly fed into the regression head.

The ablation results are summarized in Fig. 5. The full HiLoFuseNet achieved the highest decoding performance ($R = 0.587$), significantly outperforming all ablated variants ($p < 0.01$). Removing the LFS stream resulted in a moderate performance drop to $R = 0.524$ (−10.7%), indicating that low-frequency features provide complementary information beyond HGA. In contrast, removing the HGA stream caused a substantially larger degradation to $R = 0.362$ (−38.3%), confirming that HGA serves as the primary discriminative feature for finger trajectory decoding. Eliminating the DS-Conv module also significantly reduced performance ($R = 0.422$, −28.1%), demonstrating the importance of learning spatially selective cortical representations before temporal modeling. The largest degradation was observed when removing the LSTM module ($R = 0.327$, −44.3%), highlighting that temporal dynamics are critical for continuous movement decoding. Collectively, these results confirm that HiLoFuseNet benefits from the integration of high- and low-frequency neural

features, anatomically informed spatial filtering, and explicit temporal sequence modeling.

Fig. 6 illustrates the influence of two key architectural parameters: the number of spatial filters (D) in the DS-Conv module and the hidden state size (H) of the LSTM module. As the model capacity increased from small configurations ($D = 4$ or 8), decoding performance improved consistently across all tested hidden dimensions. However, beyond approximately $D = 16$, the performance gain gradually saturated. A similar trend was observed for the LSTM hidden dimension, where performance improved substantially from $H = 64$ to $H = 128$, while further increasing H yielded only marginal improvements. We selected $D = 20$ and $H = 256$ for all experiments. Importantly, this configuration balances decoding performance and model compactness. Although subject-specific hyperparameter optimization may further improve performance, a unified parameter configuration was adopted to ensure fair cross-subject and cross-dataset evaluation.

D. Computational Efficiency

Table II compares the computational cost of different feature extraction strategies using 64-channel ECoG signals with a duration of 1 second sampled at 1 kHz. All methods were evaluated in a single-threaded preprocessing environment over 1,000 simulation iterations. Compared with conventional feature extraction approaches, the proposed parallel HGA and LFS streams achieved the lowest computational latency, requiring only 2.411 ± 0.295 ms of CPU time. This represents more than a $5\times$ speedup compared with physiological band extraction and a $12\times$ speedup over Morlet wavelet extraction. These results demonstrate that the proposed feature representation not only preserves physiologically meaningful information but also substantially reduces preprocessing overhead, making it suitable for real-time neural decoding applications.

The computational efficiency of all DNN models was further evaluated in terms of floating-point operations (FLOPs), model size, and inference latency under varying channel counts (16–1024 channels), as illustrated in Fig. 7. Across all tested configurations, HiLoFuseNet maintained nearly constant model complexity, with approximately 6.720 M FLOPs and 0.341 M parameters independent of the input channel dimension, while most competing architectures exhibited substantial computation and parameter growth as channel count increased. Regarding inference latency, HiLoFuseNet demonstrated efficient real-time performance, with its mean latency scaling moderately from approximately 1.1 ms at lower channel counts to 2.6 ms at 1024 channels. In contrast, WaTEi showed progressively increasing latency as input dimensionality increased. These results demonstrate that HiLoFuseNet achieves an advantageous balance between representational capacity and computational efficiency, providing scalable and resource-efficient decoding for future high-density ECoG applications.

VI. DISCUSSION

This study demonstrates that HiLoFuseNet provides an effective and computationally efficient solution for simultaneous 5-DoF finger trajectory decoding from ECoG signals. By

TABLE II

COMPARISON OF FEATURE EXTRACTION EFFICIENCY FOR 64-CHANNEL ECoG DATA (1-SECOND DURATION, 1 KHz SAMPLING RATE). RESULTS ARE AVERAGED OVER 1,000 SIMULATION ITERATIONS.

Method	Features	CPU time (ms)
Wavelet Transform (Amplitude) [31]	Morlet wavelet amplitudes (10 frequencies, 10–150 Hz, linear scale)	29.797 ± 1.211
Wavelet Transform (Power) [32]	Morlet wavelet power (20 frequencies, 5–195 Hz, logarithmic scale)	35.472 ± 1.216
Physiological Band [23]	Spectral amplitudes in delta (1.5–5 Hz), theta (5–8 Hz), alpha (8–12 Hz), beta1 (12–24 Hz), beta2 (24–34 Hz), gamma1 (34–60 Hz), gamma2 (60–100 Hz), and gamma3 (100–130 Hz)	14.126 ± 0.681
Ours	HGA + LFS	2.411 ± 0.295

The experiment was conducted on a machine with a 13th Gen Intel(R) Core(TM) i7-13800H processor and 64 GB of RAM, running Windows 11 Home. All feature extraction was performed in a single-threaded environment using Python 3.11.5.

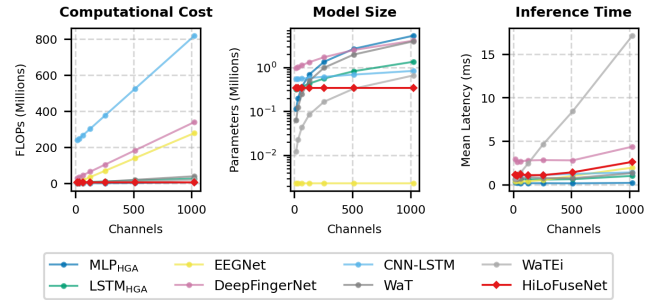


Fig. 7. Model computational efficiency comparison under varying channel counts (16, 32, 64, 128, 256, 512, and 1024 channels). Left: FLOPs (Millions). Middle: total parameters (Millions). Right: mean inference latency (ms) calculated over 1,000 simulation runs. All experiments were conducted on an NVIDIA RTX A1000 (6GB RAM).

integrating HGA and LFS within a dual-stream architecture, HiLoFuseNet achieved superior performance over representative decoding approaches, including linear regression-based methods, tensor-based models, and deep learning baselines (Fig. 2). It should be noted that direct comparisons with results reported in the literature are often confounded by differences in task settings, particularly train-test splitting protocols. To enable a fair comparison, we established a standardized benchmarking framework with causal data preprocessing and feature extraction, thereby better approximating real-time decoding scenarios in which future information is unavailable. Under the same train-test split, HiLoFuseNet achieved correlation coefficients of $R = 0.674$ on the BCIIV dataset and $R = 0.558$ on the Stanford dataset, substantially outperforming the state-of-the-art results reported by Lin et al. [32] ($R = 0.567$ on BCIIV and $R = 0.477$ on Stanford). In the following discussion, we further interpret these results from three perspectives: the role of high- and low-frequency ECoG representation, the practical advantages of simultaneous multi-finger decoding, and the remaining challenges for future clinical translation.

The high-gamma band plays a central role in ECoG-based motor decoding, as consistently reported in previous studies and reflected by the common use of high-gamma features in Table I. Our findings reinforce this view. The ablation study

(Fig. 5) showed that removing the HGA stream led to a more pronounced performance degradation than removing the LFS stream. In addition, the standalone LSTM_{HGA} model, which relied exclusively on high-gamma features, consistently ranked as the second-best method among all compared approaches (Fig. 1). These results highlight the strong predictive value of high-gamma activity for ECoG-based trajectory decoding, likely due to its close relationship with local neuronal firing rates [46]. However, HGA alone does not fully exploit the complementary information present in lower-frequency activity. Here, we used a computationally efficient LFS representation obtained by anti-aliasing and downsampling, rather than explicitly extracting multiple physiological bands such as α and β rhythms. Despite its simplicity, this coarse-grained LFS stream significantly improved decoding performance (Fig. 5), suggesting that low-frequency activity provides complementary information beyond HGA. This complementary contribution is further supported by the visualization of model attention patterns (Fig. 4). Prior studies further indicate that specific low-frequency components, especially the local motor potential, encode distinct kinematic information [17], [30]. Future extensions of HiLoFuseNet could therefore incorporate more explicit neurophysiological markers into the low-frequency stream and explore finer-grained fusion strategies. Such designs may better capture detailed kinematic information while maintaining the computational advantages demonstrated in Table II.

Dexterous hand neuroprostheses require the simultaneous decoding of multiple DoFs of finger movements from ECoG signals. Nevertheless, many prior studies on 5-DoF finger trajectory decoding have relied on independent per-finger decoders. For example, Liang and Bougrain [18], Xie *et al.* [19], Petrosyan *et al.* [29], Yao *et al.* [20], and Faes *et al.* [22] trained separate models for individual fingers. Although straightforward, this strategy requires multiple decoders to run in parallel during deployment, thereby increasing memory consumption, computational cost, and system complexity. In contrast, HiLoFuseNet adopts a unified multi-output decoding architecture that predicts all five finger trajectories within a single model. This formulation provides a more scalable alternative to per-finger decoding and is particularly suitable for neuroprosthetic implementation, where computational efficiency and low deployment complexity are critical. In addition, simultaneous prediction enables the model to learn shared neural representations across fingers and potentially capture inter-finger dependencies associated with coordinated hand movements. These properties highlight the practical advantage of HiLoFuseNet for simultaneous 5-DoF finger trajectory decoding and its potential utility in dexterous hand neuroprosthetic systems.

Despite the superior performance of HiLoFuseNet, several limitations remain. First, finger trajectory regression is still challenging, as movement and rest periods present different decoding difficulties (Fig. 3). Future work may incorporate explicit state detection [27] or post-processing gating strategies [30] to improve prediction stability during inactive periods. Second, our evaluation is limited to standard finger flexion tasks within restricted experimental sessions. To support clin-

ical translation, the proposed framework should be further evaluated under cross-session and cross-task settings. However, such validation is currently constrained by the limited availability of large-scale, standardized intracranial datasets. We therefore encourage broader Open Science efforts to share diverse ECoG datasets, similar to the Stanford library [42] used in this study, to promote the development of models that generalize across sessions, tasks, and behavioral contexts. Finally, although HiLoFuseNet was validated on standard ECoG grids and demonstrated favorable computational scalability in simulation (Fig. 7), future studies should further examine its performance on high-resolution ECoG recordings, which may offer richer spatial information for complex neural decoding tasks, such as speech decoding [47].

VII. CONCLUSION

This study presents a framework for the precise continuous decoding of finger movements from ECoG signals. The core of this approach involves a streamlined pipeline that extracts High-Gamma Activity (HGA) and Low-Frequency Signals (LFS), which are subsequently fused by our novel deep learning architecture, HiLoFuseNet. Utilizing a backbone of Depthwise Separable Convolutions and LSTM units, this dual strategy effectively addresses the feature inconsistency and computational complexity of prior methods, achieving state-of-the-art performance across two benchmark datasets. Furthermore, the model demonstrated high interpretability, exhibiting learned spatial attention patterns that align with established neurophysiological findings—specifically, focal HGA and widespread LFS. Ultimately, by providing a performant and physiologically plausible decoding solution, along with open-source benchmarks, this work is expected to advance the development of practical BCIs aimed at restoring hand function.

VIII. ACKNOWLEDGMENT

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